

# Operator Discovery Report: descriptor\_driven\_restart\_controller\_stagnation\_escape

- Operator type: restart\_controller.
- Host scaffold: nearest\_neighbor\_2opt.
- Core idea: Triggers deterministic restarts on short stagnation and grows restart budget cautiously, using perturbation at each restart to escape nearest-neighbor / 2-opt traps without excessive diversification.
- Novelty classification: adaptive\_restart\_controller.
- Rediscovery signature: restart\_controller:2:6:1.
- Surviving candidate: False.
- Problem class where it helps: two\_cluster\_bottleneck\_tsp.

## Pseudocode

- Start from a conservative restart budget.
- Increase restart count only after the configured stagnation trigger.
- Keep perturbation restarts bounded by a deterministic maximum.

## Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 586d7fcf0845f2b3.

## Why It Should Help

- The operator changes restart pressure under stagnation, which is a compact way to regulate exploration without rewriting search internals.

## Failure Cases

- Does not transplant cleanly into sparse\_three\_opt.
- Does not transplant cleanly into clustered\_local\_search.
- Runtime inflation is high at 0.911918.

## Transplant Results

- nearest\_neighbor\_2opt gap delta 0.0, runtime delta 89.435386, distance-eval delta 3270.285714.
- cheapest\_insertion\_2opt gap delta -0.000585, runtime delta 1949.318443, distance-eval delta 4254.285714.
- random\_restart\_2opt gap delta 0.0, runtime delta -235.805214, distance-eval delta -13437.714286.
- sparse\_three\_opt gap delta 0.036385, runtime delta 4.635029, distance-eval delta -65.0.
- clustered\_local\_search gap delta 0.003398, runtime delta -29.1608, distance-eval delta -4644.285714.

## Instance-Family Results

- heldout\_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered\_tsp operator gap 0.110431 vs baseline 0.110853.
- elongated\_corridor\_tsp operator gap 0.003204 vs baseline 0.003204.
- grid\_like\_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest\_neighbor\_trap\_tsp operator gap 0.0 vs baseline 0.0.
- two\_cluster\_bottleneck\_tsp operator gap 0.084632 vs baseline 0.109057.
- uniform\_euclidean operator gap 0.0 vs baseline 0.0.

## **Pareto Results**

- Gap delta: -0.003304.
- Runtime inflation: 0.911918.
- Distance-evaluation inflation: 0.929896.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

## **Ablation Results**

- disabled mean gap 0.121317 and delta vs full operator 0.003304.
- fixed\_restart\_budget mean gap 0.118013 and delta vs full operator 0.0.
- wider\_restart\_budget mean gap 0.117688 and delta vs full operator -0.000325.